

Topic: Recursive Queries

Q. What are Recursive Queries in SQL? Explain the concept of Common Table Expressions (CTE) and how a Recursive CTE works with an example.

> **Definition to Remember**

> A **Recursive CTE** is used to query hierarchical data or tree structure. It is used to query data that is recursive in nature. It is used to query data that is recursive in nature. It is used to query data that is recursive in nature.

Recursive Queries & Common Table Expressions (CTE)

A **CTE** is a temporary, named result set defined using the `WITH` keyword. It exists only during the execution of a single SQL statement.

- Use

2. Structure of a Recursive CTE

A recursive CTE has exactly **3 components**:

ANCHOR MEMBER Base case (executes once, gives R)

WITH

SELEC

UNION ALL Connects anchor + recursive

RECURSIVE MEMBER References CTE itself (gives R, R, ...)

SELEC

)

SELEC

3. Working of a Recursive Query

1. **Anchor Member** executes first produces base result set R₀.

2. **Recursive Member** executes repeatedly until it reaches the base case.

4. Example Organizational Hierarchy

Table: `Employee (EmpID, Name, ManagerID)` Find all levels of management.

```sql

UNION ALL

-- Recursive: Find employees managed by those already found

SELEC

## 5. Applications

- **Organizational Charts:** Employee hierarchy traversal.

- **B**

---

> **Must-Write Points (for 10 marks)**

> 1. R

---

> **Quick Recall**

> `Rec  
when e